

On predator-prey systems and small-gain theorems*

Patrick De Leenheer[†], David Angeli[‡] and Eduardo D. Sontag[§]

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Abstract

This paper deals with an almost global attractivity result for Lotka-Volterra systems with predator-prey interactions. These systems can be written as (negative) feedback systems. The subsystems of the feedback loop are monotone control systems, possessing particular input-output properties. We use a small-gain theorem, adapted to a context of systems with multiple equilibrium points to obtain the desired almost global attractivity result. It provides sufficient conditions to rule out oscillatory or more complicated behavior which is often observed in predator-prey systems.

1 Introduction

Predator-prey systems have been -and still are- attracting a lot of attention [8, 18, 11, 5] since the early work of Lotka and Volterra. It is well-known that these systems may exhibit oscillatory behavior, the best known example being the classical Lotka-Volterra predator-prey system, see e.g. [8, 9], defined by

$$\begin{pmatrix} \dot{x} \\ \dot{z} \end{pmatrix} = \text{diag}(x, z) \left(\begin{pmatrix} 0 & +a_{12} \\ -a_{21} & 0 \end{pmatrix} \begin{pmatrix} x \\ z \end{pmatrix} + \begin{pmatrix} -r_1 \\ r_2 \end{pmatrix} \right)$$

where x and z denote the predator, respectively the prey concentrations and a_{12}, a_{21}, r_1 and r_2 are positive constants. The phase portrait consists of an infinite number of periodic solutions centered around an equilibrium point. It is also well-known that this system is not structurally stable. In fact, it has been shown in [9] that the more general, but not necessarily predator-prey type, Lotka-Volterra system:

$$\begin{pmatrix} \dot{x} \\ \dot{z} \end{pmatrix} = \text{diag}(x, z) \left(\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ z \end{pmatrix} + \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \right)$$

where there is no restriction on the signs of the parameters a_{ij} , r_k , does not exhibit nontrivial *isolated* periodic solutions. Hence, compelling evidence of oscillatory behavior in predator-prey systems is not provided by the classical Lotka-Volterra predator-prey system or by any 2-dimensional Lotka-Volterra system. But as we will see next oscillations can be found in different predator-prey models.

One predator-prey system (which is still low-dimensional but not of the Lotka-Volterra type) is Gause's model [9] which may admit isolated periodic solutions under suitable conditions [11]:

$$\begin{aligned} \dot{x} &= x(q(z) - d) \\ \dot{z} &= zg(z) - xp(z) \end{aligned} \tag{1}$$

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[†]corresponding author: Arizona State University, Department of Mathematics and Statistics, Tempe, AZ 85287, email: leenheer@math.la.asu.edu

[‡]Dip. di Sistemi e Informatica Università di Firenze, Via di S. Marta 3, 50139 Firenze, Italy, email: angeli@dsi.unifi.it

[§]Supported in part by US Air Force Grant F49620-01-1-006, NIH grant P20 GM64375, and by the Rutgers Institute for Biology at the Interface of the Mathematical and Physical Sciences (BioMaPS Institute). Department of Mathematics, Rutgers University, New Brunswick, NJ 08903, <http://www.math.rutgers.edu/~sontag>, email: sontag@control.rutgers.edu

where $g(z)$ is the growth rate of the prey in absence of the predator (often $zg(z)$ is logistic) and $p(z)$ is the so-called predator functional response, a non-negative, increasing function which is zero at zero (often of Michaelis-Menten type). If the function $q(z)$ is proportional to $p(z)$ -as is usually the case- then the proportionality factor is called the conversion rate. Finally, $d > 0$ is the death rate of predators.

Oscillatory behavior can also be found *within* the class of Lotka-Volterra predator-prey systems, but then the number of predator and prey species must be larger than two. As an illustration we will provide two examples with 2 predator species and 1 prey species. A common property for these examples is that the predator species are *mutualistic*, i.e. the effect of one predator on another is not negative. This might occur if the predator population is stage-structured, for instance if it consists of immature and mature predator species.

Example 1

Consider the parameterized (parameter $k > 0$) Lotka-Volterra predator-prey system with 2 predator species x_1 and x_2 and 1 prey species z :

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{z} \end{pmatrix} = \text{diag}(x_1, x_2, z) \left(\begin{pmatrix} -1 & 1 & 1 \\ 1 & -2 & 0 \\ 0 & -k & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ z \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ k+3 \end{pmatrix} \right) \quad (2)$$

Suppose that x_1 is interpreted as the immature and x_2 as the mature predators. Notice that only the mature predator kills the prey but does not consume it. The immature predators on the other hand, do not kill but do consume the prey. This means that the predator population consists of adults who hunt for food for their young but do not eat the prey themselves. Obviously the mature predator must find food elsewhere and this is reflected in the $x_2 \cdot (+1)$ term in the $\dot{x}_2$ - equation. For every $k > 0$ there is a nontrivial equilibrium point at $(1, 1, 1)$ and a simple application of the Routh-Hurwitz criterion reveals that this equilibrium point is locally asymptotically stable if $k \in (0, k_c)$ where $k_c := 57$. For $k > k_c$ however, the linearization at $(1, 1, 1)$ possesses 1 stable (and hence real) eigenvalue and 2 unstable eigenvalues. It can be shown that for $k - k_c > 0$ but small, the unstable eigenvalues must be complex conjugate with nontrivial imaginary part. This suggests the occurrence of a Hopf bifurcation at the critical value k_c . In fact we determined the occurrence of a supercritical Hopf bifurcation and hence the existence of stable oscillatory behavior for system (2), see figure 1. To establish this we used the method outlined in [6] on p. 153 and the software Maple to perform some of the calculations. We briefly sketch the involved steps. First, a simple translation to the equilibrium point $(1, 1, 1)^T$ is performed by setting $z = x - (1, 1, 1)^T$. In the new z -coordinates the equations are $\dot{z} = A(k)z + \text{diag}(z)A(k)z$, where $A(k)$ is the interaction matrix corresponding to the Lotka-Volterra system (2). To obtain the desired result we need to show that for the z -system two conditions from the Hopf bifurcation theorem are satisfied when k passes through the critical value $k_c = 57$. The first, transversality condition of the Hopf bifurcation theorem (which expresses that the complex pair of eigenvalues of $A(k)$ crosses the imaginary axis at k_c) is easily verified using the information from the Routh-Hurwitz criterion and the relationship between eigenvalues of $A(k)$ and the numbers from a Routh-Hurwitz table. The second condition requires the calculation of a number a associated to the z -system at the critical value k_c , see (3.4.11) in [6]. From now on fix k at k_c and denote $A(k_c)$ by A . A linear transformation $z = TX$ is performed to put A in block-diagonal form $\hat{A}$ where the upper 2×2 -block corresponds to the imaginary eigenvalue pair $\{\pm\sqrt{10}i\}$ and the lower 1×1 -block corresponds to the stable eigenvalue -6 . This leads to $\dot{X} = \hat{A}X + T^{-1} \text{diag}(TX)AX$. The problem now is that an approximation of the center manifold for this system to *at least quadratic terms* should be computed to verify the mentioned second condition. To compute this approximation we used Maple. This approximation is in turn used to calculate the number a , see (3.4.11) in [6]. For this particular example a turns out to be negative which allows us to conclude that a supercritical Hopf bifurcation occurs.

Example 2

Consider the parameterized (parameter $k > 0$) Lotka-Volterra predator-prey systems with 2 predators and 1 prey species:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{z} \end{pmatrix} = \text{diag}(x_1, x_2, z) \left(\begin{pmatrix} -1 & \frac{7}{2} & \frac{1}{2} \\ 0 & -2 & 1 \\ -k & -\frac{k}{2} & -\frac{3}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ z \end{pmatrix} + \begin{pmatrix} -3 \\ 1 \\ \frac{3}{2}(k+1) \end{pmatrix} \right) \quad (3)$$

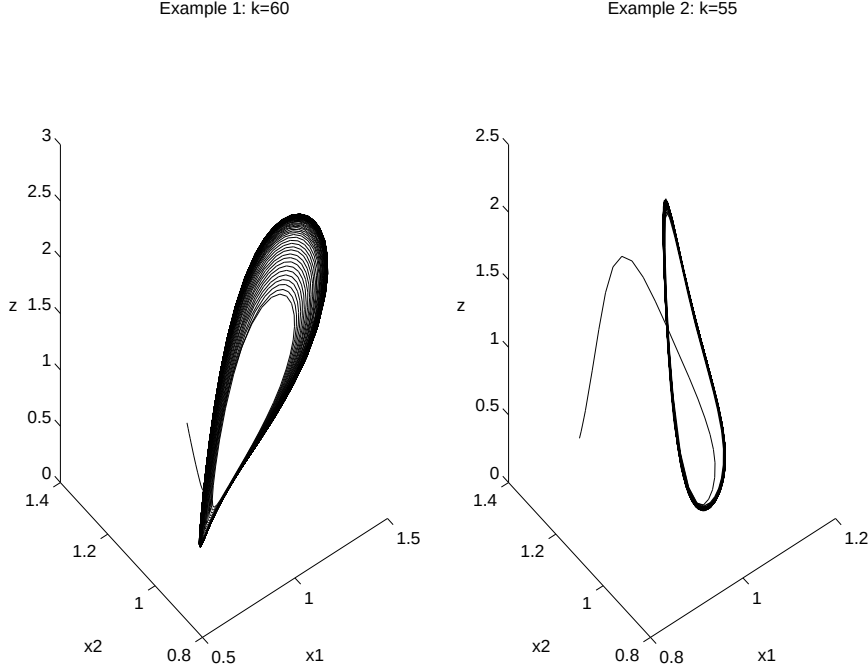


Figure 1: Oscillations in Examples 1 and 2 (initial condition $(0.8, 1.1, 0.9)$, integration over $[0, 200]$).

As in example 1, $(1, 1, 1)$ is always an equilibrium point and it can be shown that a supercritical Hopf bifurcation occurs at $k = k_c$ where $k_c = 105/2 = 52.5$, again illustrating (stable) oscillatory behavior, see figure 1.

Equipped with convincing evidence for possible oscillations in 3-dimensional Lotka-Volterra predator-prey systems, it might be expected that this or even more complicated behavior is possible in the following, in general higher-dimensional, Lotka-Volterra predator-prey system, which will be the primary focus of attention in this paper:

$$\begin{pmatrix} \dot{x} \\ \dots \\ \dot{z} \end{pmatrix} = \text{diag}(x, z) \begin{pmatrix} A & \vdots & B \\ \dots & \dots & \dots \\ -C & \vdots & D \end{pmatrix} \begin{pmatrix} x \\ \dots \\ z \end{pmatrix} + \begin{pmatrix} r_1 \\ \dots \\ r_2 \end{pmatrix} \quad (4)$$

where x is k -dimensional and z is $(n-k)$ -dimensional. Throughout this paper we make the following assumption:

H: For system (4), A and D are Metzler and stable and $B, C \geq 0$

where the inequalities on the matrices B and C should be interpreted entry-wise. Recall that a matrix is a Metzler matrix if its off-diagonal entries are non-negative. A matrix is stable if it only has eigenvalues with negative real part. Examples 1 and 2 satisfy these properties.

Some remarks concerning system (4) are in order here:

1. System (4) is a Lotka-Volterra predator-prey system consisting of k predator species x and $(n-k)$ prey species z . The interaction within both sub-communities is mutualistic. Especially for the predator sub-community this differs from the usual assumption that the interaction between predators is competitive.
2. There are no restrictions (nor will any restrictions be introduced later in the paper) on the signs of the components of r_1 and r_2 . These components are the death or growth rates of the species that do not originate from the interaction with the other species.

In this paper we consider whether oscillations or more complicated behavior of system (4) can be ruled out. In fact, we are mainly interested in the more restrictive problem of finding conditions for

the existence of an (almost) globally asymptotically stable equilibrium point. In view of Examples 1 and 2 this is a nontrivial problem.

It is well-known that in general, Lotka-Volterra systems may display complicated behavior, ranging from oscillatory behavior, over heteroclinic cycles to chaos. A useful reference in this respect is [9], especially for the many references it contains. For results on competitive systems we refer to [12, 3] and for predator-prey systems to [18]. In the latter reference however (and also in other work on predator-prey systems), the assumption is usually that when in isolation, the predator populations and the prey populations interact competitively. This is different from our assumption that they interact in a mutualistic way.

We also point out that there exists an extensive literature [17, 14, 19, 20, 21] on the related class of systems consisting of two competing sub-communities of mutualists. If Lotka-Volterra interactions are assumed, these systems are given by the following equations:

$$\begin{pmatrix} \dot{x} \\ \dots \\ \dot{z} \end{pmatrix} = \text{diag}(x, z) \begin{pmatrix} A & \vdots & -P \\ \dots & \dots & \dots \\ -Q & \vdots & D \end{pmatrix} \begin{pmatrix} x \\ \dots \\ z \end{pmatrix} + \begin{pmatrix} r_1 \\ \dots \\ r_2 \end{pmatrix} \quad (5)$$

where A, D are Metzler and stable and $P, Q \geq 0$. The mentioned references are devoted to global stability properties of equilibria and to persistence. Remarkable results have been obtained for this class. But we emphasize that there is a fundamental difference between system (5) and our system (4). Indeed, the flow of system (5) is monotone [7, 13], while the flow of our model (4) does not possess this property. It is well-known that monotonicity properties are often useful in establishing convergence to and stability of equilibria. The lack of a monotonicity property for system (4) forces one to use different tools to prove (almost) global stability of equilibria. We believe that the perspective of control systems might be useful in achieving this. We elaborate briefly on this claim next.

To system (4) one can associate two Input/Output (I/O) systems:

$$\begin{aligned} \dot{z} &= \text{diag}(z)(Dz + r_2 + Cu(t)) \\ w &= z \end{aligned} \quad (6)$$

and

$$\begin{aligned} \dot{x} &= \text{diag}(x)(Ax + r_1 + Bv(t)) \\ y &= x \end{aligned} \quad (7)$$

where $u(t)$ is a (component-wise) non-positive and $v(t)$ a (component-wise) non-negative input signal and w and y are output signals. These I/O-systems *are* monotone in the sense of [1] (we shall provide a precise definition of monotone I/O systems in a later section). Associated to both these I/O systems are what we termed *I/O quasi-characteristics* k_w , respectively k_y (see Definition 2). Loosely speaking, such a characteristic is a mapping between the input and output space capturing the ability of a certain I/O system to convert a constant input into a converging output where the limit is (almost) independent of initial conditions. The I/O quasi-characteristic assigns to every input its corresponding output limit.

Notice that system (4) can easily be identified as the *negative feedback interconnection* of system (6) and system (7), see figure 2, by setting:

$$\begin{aligned} v &= w \\ u &= -y. \end{aligned} \quad (8)$$

The fact that system (4) is a feedback interconnection of two systems, opens up the toolbox from the theory of interconnected control systems to prove global stability. One particular tool we will use is a so-called ‘small-gain theorem’. An informal statement of our main result is the following.

Theorem 1. *If the discrete-time system*

$$u_{k+1} = -(k_y \circ k_w)(u_k)$$

possesses a globally attractive fixed point, then the feedback system (6), (7) and (8), or equivalently system (4) possesses an (almost) globally attractive equilibrium point.

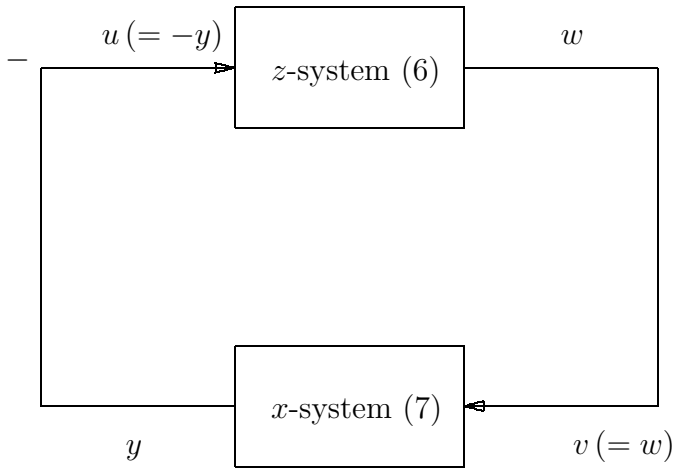


Figure 2: Feedback interconnection

As an illustration of this result we will provide sufficient conditions for the gain k in Examples 1 and 2, guaranteeing that the condition of theorem 1 is satisfied.

The development of a theory for monotone control systems has been initiated in [1]. A particular small-gain theorem has been proved there, but it is not applicable in our context. An appropriate extension is given in [2] however and this allows us to formulate sufficient conditions for the existence of an almost globally attractive equilibrium point of system (4). Note that almost global attractivity of an equilibrium point is in a sense the strongest achievable stability property for a Lotka-Volterra system. Indeed, these systems typically possess multiple equilibrium points. (for instance, zero is *always* an equilibrium point; of course it is usually an uninteresting one from biological point of view)

2 Preliminaries

2.1 Monotone I/O systems and a small gain theorem

The material in this section can be found in a far more general setting in [1, 2]. We restrict to a framework that serves our purposes, namely I/O systems described by differential equations. Consider the following I/O system:

$$\begin{aligned}\dot{x} &= f(x, u) \\ y &= h(x)\end{aligned}\tag{9}$$

where $x \in \mathbb{R}^n$ is the state, $u \in U \subset \mathbb{R}^m$ the input and $y \in Y \subset \mathbb{R}^p$ the output. It is assumed that f and g are smooth (say continuously differentiable) and that the input signals $u(t) : \mathbb{R} \rightarrow U$ are Lebesgue measurable functions and locally essentially bounded (i.e. for every compact time interval $[T_m, T_M]$, there is some compact set C such that $u(t) \in C$ for almost all $t \in [T_m, T_M]$). This implies that solutions with initial states $x_0 \in \mathbb{R}^n$ are defined for all inputs $u(\cdot)$ and will be denoted by $x(t, x_0, u(\cdot))$, $t \in \mathcal{I}$ where $\mathcal{I}$ is the maximal interval of existence for this solution. See [15] for a general theoretical framework for the analysis of I/O systems. From now on we will assume that a fixed set $X \subset \mathbb{R}^n$ is given which is forward invariant, i.e. for all inputs $u(\cdot)$ and for every $x_0 \in X$ it holds that $x(t, x_0, u(\cdot)) \in X$, for all $t \in \mathcal{I} \cap \mathbb{R}_+$. Henceforth initial conditions are restricted to this set X . We will be particularly interested in cases where $X = \mathbb{R}_+^n$, $U = \mathbb{R}_+^m$ or $U = -\mathbb{R}_+^m$.

We denote the usual partial order on $\mathbb{R}^n$ by $\preceq$, i.e. for $x, y \in \mathbb{R}^n$, $x \preceq y$ means that $x_i \leq y_i$ for $i = 1, \dots, n$. The state space X (input space U , output space Y) inherits the partial order from $\mathbb{R}^n$ ($\mathbb{R}^m$, $\mathbb{R}^p$) as the former sets are subsets of the latter ones. Similarly, the partial order on $\mathbb{R}^m$ carries over to the set of input signals in a natural way (hence we use the same notation for the partial order on this latter set): $u(\cdot) \preceq v(\cdot)$ if $u(t) \preceq v(t)$ for almost all $t \geq 0$. The next

definition introduces the concept of a monotone I/O system which, loosely speaking means that ordered initial conditions and input signals lead to subsequent ordered solutions.

Definition 1. *The I/O system (9) is monotone (with respect to the usual partial orders) if the following conditions hold:*

$$x_1 \preceq x_2 \text{ and } u(\cdot) \preceq v(\cdot) \Rightarrow x(t, x_1, u(\cdot)) \preceq x(t, x_2, v(\cdot)) \text{ for all } t \in (\mathcal{I}_1 \cap \mathcal{I}_2) \cap \mathbb{R}_+. \quad (10)$$

and

$$h \text{ is a monotone map, i.e. } x_1 \preceq x_2 \Rightarrow h(x_1) \preceq h(x_2). \quad (11)$$

Remark 1. We refer to [1] for tests to check whether a given I/O system is monotone.

Remark 2. Since no confusion about the partial orders on input, state and output spaces is possible here (we always mean the usual partial order $\preceq$) we will in the sequel refer to *monotone I/O systems* and not explicitly mention the involved partial orders. However we emphasize that in general a the concept of a monotone I/O system requires the enumeration of these partial orders, see [1].

Of particular interest is how an I/O system behaves when it is supplied with a *constant* input. Next we introduce a notion which implies that this behavior is fairly simple [2].

Definition 2. *Assume that X has full measure. The I/O system (9) possesses an Input/State (I/S) quasi-characteristic $k_x : U \rightarrow X$ if for every constant input $u \in U$ (and using the same notation for the corresponding $u(\cdot)$), there exists a set of measure zero $\mathcal{B}_u$ such that:*

$$\forall x_0 \in X \setminus \mathcal{B}_u : \lim_{t \rightarrow +\infty} x(t, x_0, u) = k_x(u) \quad (12)$$

If system (9) possesses an I/S quasi-characteristic k_x then it also possesses an Input/Output (I/O) quasi-characteristic $k_y : U \rightarrow Y$ defined as $k_y := h \circ k_x$.

Remark 3. An important property of a static I/S or I/O quasi-characteristic of a *monotone* I/O system is that it is a **monotone map**. Indeed, for any pair of constant inputs $u \preceq v$ one may find an initial condition $x_0 \in X \setminus (\mathcal{B}_u \cup \mathcal{B}_v)$ such that (10) is satisfied when choosing $x_1 = x_2 = x_0$. Upon taking limits for $t \rightarrow +\infty$ of both sides of the last inequality in (10) and using (12), we see that k_x is monotone. The same is true for an I/O quasi-characteristic k_y since the output map is monotone by (11) and the composition of monotone maps is monotone.

We are ready to state the main tool in proving stability for Lotka-Volterra predator-prey systems. This is a special case of a more general result proved in [2]. Below we use the concept of an *almost globally attractive equilibrium point* of an autonomous system, which means that there exists an equilibrium point of this system which attracts all solutions which are not initiated in a certain set of measure zero. Similarly, an *almost globally asymptotically stable equilibrium point* is an equilibrium point which is stable (in the Lyapunov sense) and almost globally attractive.

Theorem 2. *Consider the following two I/O systems:*

$$\dot{x}_1 = f_1(x_1, u_1), \quad y_1 = h_1(x_1) \quad (13)$$

$$\dot{x}_2 = f_2(x_2, u_2), \quad y_2 = h_2(x_2) \quad (14)$$

where $x_i \in X_i \subset \mathbb{R}^{n_i}$, $u_i \in U_i \subset \mathbb{R}^{m_i}$ and $y_i \in Y_i \subset \mathbb{R}^{p_i}$ for $i = 1, 2$. Suppose that $Y_1 = U_2$ and $Y_2 = -U_1$ and that the I/O systems are interconnected through a (negative) feedback loop:

$$u_2 = y_1 \quad (15)$$

$$u_1 = -y_2. \quad (16)$$

Assume that:

1. *Both I/O systems (13) and (14) are monotone.*
2. *Both I/O systems (13) and (14) possess continuous I/S quasi-characteristics k_{x_1} and k_{x_2} respectively (and thus also I/O quasi-characteristics k_{y_1} and k_{y_2}).*
3. *All forward solutions of the feedback system (13) – (16) are bounded.*

Then the feedback system possesses an almost globally attractive equilibrium point $(\bar{x}_1, \bar{x}_2) \in X_1 \times X_2$ if the following discrete-time system, defined on U_1 :

$$u_{k+1} = -(k_{y_2} \circ k_{y_1})(u_k) \quad (17)$$

possesses a globally attractive fixed point $\bar{u} \in U_1$. In that case $(\bar{x}_1, \bar{x}_2) = (k_{x_1}(\bar{u}), (k_{x_2} \circ k_{y_1})(\bar{u}))$.

This result and similar ones following later in the paper, are called *small gain theorems*. The last condition is often referred to as a *small gain condition*. We will use this terminology in the sequel. (For another example of application of small-gain ideas in biology, see [16]).

2.2 Boundedness and stability of Lotka-Volterra systems

Consider the classical Lotka-Volterra system:

$$\dot{x} = \text{diag}(x)(Ax + r) \quad (18)$$

where $x \in \mathbb{R}^n$ and $r \in \mathbb{R}^n$. Note that there are no assumptions on the sign of the entries of A or the components of r . It is possible to show that $\mathbb{R}_+^n$ is a forward invariant set for (18), see e.g. Theorem 3 in [1] and also the next subsection for a more general result on forward invariance of $\mathbb{R}_+^n$ of I/O Lotka-Volterra systems. In the sequel we will therefore assume that initial conditions are restricted to $\mathbb{R}_+^n$. The following result characterizes *uniformly bounded* Lotka-Volterra systems [9]. Recall that a Lotka-Volterra system is uniformly bounded if there exists a compact, absorbing set $K \subset \mathbb{R}_+^n$, i.e. for all $x_0 \in \mathbb{R}_+^n$, there is a $T(x_0) \geq 0$ such that $x(t, x_0) \in K$ for all $t \geq T(x_0)$. Below we use the notation $\text{int}(\mathbb{R}_+^n)$ for the interior points of $\mathbb{R}_+^n$ (i.e. those vectors in $\mathbb{R}_+^n$ having only strictly positive components).

Lemma 1. (*Exercise 15.2.7, p.188 in [9]*) *System (18) is uniformly bounded if and only if*

$$\exists c \in \text{int}(\mathbb{R}_+^n) : -Ac \in \text{int}(\mathbb{R}_+^n). \quad (19)$$

and every principal sub-matrix of A has the same property.

Matrices satisfying condition (19) are known as *B-matrices*.

We will now specialize to Lotka-Volterra systems with an interaction matrix A which is Metzler. But first we collect some well-known facts about the stability of Metzler matrices (see e.g. [9]). They are consequences of the Perron-Frobenius Theorem, see e.g. [13, 9].

Lemma 2. (*Theorem 15.1.1, p.181 in [9]*) *A Metzler matrix is stable if and only if it is diagonally dominant, i.e.*

$$\exists d \in \text{int}(\mathbb{R}_+^n) : -Ad \in \text{int}(\mathbb{R}_+^n). \quad (20)$$

If A is a stable Metzler matrix then (20) obviously also holds for every principal sub-matrix of A , implying that every principal sub-matrix of A is also stable. In other words, a Metzler matrix is stable if and only if it is a B-matrix.

The following result is an immediate application of results in [17, 9, 10]. The support set of $x \in \mathbb{R}_+^n$ is defined as $\text{supp}(x) := \{y \in \mathbb{R}_+^n \mid y_i > 0 \text{ if } x_i > 0\}$.

Lemma 3. (*Theorem 15.3.1, p.191 in [9]*) *If A is a stable Metzler matrix, then system (18) possesses a unique equilibrium point $\bar{x}$ which is globally asymptotically stable with respect to initial conditions in its support set $\text{supp}(\bar{x})$. Suppose that x^e is an equilibrium point of (18). Then x^e is globally asymptotically stable with respect to initial conditions in $\text{supp}(x^e)$ (and hence $x^e = \bar{x}$) if and only if the following condition is satisfied:*

$$Ax^e + r \leq 0 \quad (21)$$

Remark 4. For future reference we provide an explicit characterization of an arbitrary equilibrium point $x^e \in \mathbb{R}_+^n$ (which is not necessarily $\bar{x}$ from the above lemma) of system (18) in case A is a stable Metzler matrix. If $x^e \in \mathbb{R}_+^n$ is an equilibrium point of system (18), then there exists a partition I, J of the index set $N := \{1, 2, \dots, n\}$ (i.e. $N = I \cup J$ and $I \cap J = \emptyset$ where one of the sets I or J could be empty) such that $x_i^e = 0$ for $i \in I$ and $x_j^e > 0$ for $j \in J$. This implies that for

all $j \in J$, the j -th component of the vector $Ax^e + r$ must be 0. Equivalently, denoting the vector (x_j^e) , $j \in J$ by x_s^e , there must exist a principal sub-matrix A_s of A (which is also stable and hence invertible by Lemma 2) such that:

$$x_s^e = -A_s^{-1}r_s$$

where r_s is obtained from r by deleting all components r_i with $i \in I$.

We are now in a position to prove a boundedness result for the system of interest (4).

Lemma 4. *The solutions of system (4) are uniformly bounded provided $\mathbf{H}$ holds.*

Proof. By lemma 1 it suffices to show that the matrix

$$\tilde{A} = \begin{pmatrix} A & \vdots & B \\ \dots & \dots & \dots \\ -C & \vdots & D \end{pmatrix}$$

is a B-matrix or equivalently, that $\tilde{A}$ and all its principal sub-matrices satisfy condition (19). Since A and D are stable Metzler matrices by $\mathbf{H}$, it follows from lemma 2 that there exists $d_1 \in \text{int}(\mathbb{R}_+^k)$ and $d_2 \in \text{int}(\mathbb{R}_+^{(n-k)})$ such that $-Ad_1 \in \text{int}(\mathbb{R}_+^k)$ and $-Dd_2 \in \text{int}(\mathbb{R}_+^{(n-k)})$. Since $B, C \geq 0$ by $\mathbf{H}$, there exists a sufficiently large real number $\alpha > 0$ such that:

$$\tilde{A} \begin{pmatrix} \alpha d_1 \\ d_2 \end{pmatrix} = \begin{pmatrix} A & \vdots & B \\ \dots & \dots & \dots \\ -C & \vdots & D \end{pmatrix} \begin{pmatrix} \alpha d_1 \\ d_2 \end{pmatrix} \in -\text{int}(\mathbb{R}_+^n)$$

Since by lemma 2, the principal sub-matrices of A and D are also Metzler and stable and therefore also satisfy the diagonally dominance condition (20), the same argument we used to prove that $\tilde{A}$ satisfies (19), can be used to prove that all principal sub-matrices of $\tilde{A}$ also satisfy (19). This concludes the proof. $\square$

2.3 Lotka-Volterra systems with inputs

Consider a classical Lotka-Volterra system subject to an input:

$$\dot{x} = \text{diag}(x)(Ax + r + Bu) \quad (22)$$

where $x \in \mathbb{R}^n$, $u \in U$ is the input. We assume that $U = \mathbb{R}_+^m$ or $U = -\mathbb{R}_+^m$. The input signals $u(\cdot) : \mathbb{R} \rightarrow U$ are Lebesgue measurable and locally essentially bounded functions. Note that there is no assumption on the sign of the entries of B nor on the components of r . It can be shown that $\mathbb{R}_+^n$ is forward invariant. This follows from an application of Theorem 3 in [1]. To apply this result we first denote the right-hand side of (22) as $f(x, u)$ and observe that f is locally Lipschitz in x , locally uniformly in u . Secondly, denoting $f_D(x) := \{f(x, u) \mid u \in D\}$ where D is an arbitrary compact subset of U , we need to verify whether

$$\forall x \in \mathbb{R}_+^n : f_D(x) \subset T_x \mathbb{R}_+^n$$

holds, where $T_x \mathbb{R}_+^n$ is the tangent cone to $\mathbb{R}_+^n$ at $x \in \mathbb{R}_+^n$. This cone is defined as follows:

$$T_x \mathbb{R}_+^n := \left\{ \lim_{h_i \searrow 0} \frac{1}{h_i} (y_i - x) \mid y_i \rightarrow x \text{ while } y_i \in \mathbb{R}_+^n \text{ and } h_i > 0 \text{ as } i \rightarrow +\infty \right\}$$

This second condition is also easily verified, yielding that $\mathbb{R}_+^n$ is forward invariant for system (22). Hence in the sequel we will always restrict initial conditions to $\mathbb{R}_+^n$.

Lemma 5. *If A is a stable Metzler matrix, then system (22) possesses a continuous I/S quasi-characteristic $k_x : U \rightarrow \mathbb{R}_+^n$.*

Proof. Step 1: Existence of k_x

This follows immediately from the first part of lemma 3. Denote the stable equilibrium point corresponding to an arbitrary $u \in U$ as $k_x(u)$. Then the set $\mathcal{B}_u$ of non-converging initial conditions is $\mathbb{R}_+^n \setminus \text{supp}(k_x(u))$ which is a subset of the boundary of $\mathbb{R}_+^n$. Clearly, $\mathcal{B}_u$ is of measure zero.

Step 2: Continuity of k_x

To prove continuity of k_x it is sufficient to show that k_x is a locally bounded function (i.e. for every compact set $C \subset U$, $k_x(C)$ is a bounded set) and that the graph of k_x is a closed set. By lemma 3 and remark 4 we know that to every $u \in U$ corresponds a unique equilibrium point $k_x(u)$ for which the vector $[k_x(u)]_s$ of nonzero components can be explicitly characterized as

$$[k_x(u)]_s = -A_s^{-1}(u)(r_s(u) + [Bu]_s)$$

where $r_s(u)$ and $[Bu]_s$ are obtained from r respectively Bu by deleting those components corresponding to zero components of $k_x(u)$. Note the explicit dependence of A_s , r_s and $[Bu]_s$ on u . Local boundedness of $k_x(u)$ will follow from the following chain of (in)equalities, where $|\cdot|$ denotes any norm on $\mathbb{R}^n$ (or on a lower dimensional space $\mathbb{R}^l$ with $l < n$) and $\|\cdot\|$ stands for its associated matrix norm.

$$\begin{aligned} |k_x(u)| &= |[k_x(u)]_s| \\ &= |-A_s^{-1}(u) \cdot (r_s(u) + [Bu]_s)| \\ &\leq \| -A_s^{-1}(u) \| \cdot (|r_s(u)| + |[Bu]_s|) \\ &\leq \left\| \int_0^\infty e^{A_s(u)t} dt \right\| \cdot (|r| + |Bu|) \\ &\leq \int_0^\infty \| e^{A_s(u)t} \| dt \cdot (|r| + \|B\| \cdot |u|) \\ &\leq \int_0^\infty M_s e^{\lambda_F(A_s(u))t} dt \cdot (|r| + \|B\| \cdot |u|) \\ &\leq \frac{M_s}{-\lambda_F(A_s(u))} \cdot (|r| + \|B\| \cdot |u|) \\ &\leq \frac{M}{-\lambda_F(A)} \cdot (|r| + \|B\| \cdot |u|) \end{aligned}$$

where we denoted the dominating Perron-Frobenius eigenvalue of a Metzler matrix P , see e.g. [4], by $\lambda_F(P)$, $M := \max(M_s)$ (see item 2 below for the definition of M_s ; note that $\max(M_s)$ exists since there are only a finite number of principal sub-matrices and hence only a finite number of M_s 's) and where we used the following facts:

1. In the 4-th step we used the identity $-P^{-1} = \int_0^\infty e^{Pt} dt$ for any stable matrix P **and** the fact that all principal sub-matrices of a stable Metzler matrix are stable, see Lemma 3 (this last fact is also used in the 7-th step when performing the integration).
2. For any Metzler matrix P_s , $\|e^{P_s t}\| \leq M_s e^{\lambda_F(P_s)t}$ for some constant $M_s > 0$ in the 6-th step.
3. $\lambda_F(P_s) \leq \lambda_F(P)$ for any principal sub-matrix P_s of a Metzler matrix P (this follows from an immediate application of Corollary 1.6 in [4]) in the last step.

Next we will prove that the graph of k_x :

$$\text{graph}(k_x) := \{(u, x) \in U \times \mathbb{R}_+^n | x = k_x(u)\}$$

is a closed set in the topology $U \times \mathbb{R}_+^n$. (recall that $U = \mathbb{R}_+^m$ or $U = -\mathbb{R}_+^m$)

Define:

$$V := \{(u, y) \in U \times \mathbb{R}_+^n | Ay + r + Bu \leq 0 \text{ and } y^T(Ay + r + Bu) = 0\}$$

Clearly V is a closed set with respect to the subspace topology. Now it follows from lemma 3 and the particular form of (22) that $\text{graph}(k_x) = V$, so $\text{graph}(k_x)$ is closed also and this concludes the proof. $\square$

2.4 Global asymptotic stability of fixed points of scalar non-increasing maps

In this subsection we collect some results for checking global asymptotic stability of fixed points of discrete-time systems satisfying a particular condition. They are useful when verifying the small-gain condition which appears in our main results in the next section.

Consider the following scalar discrete-time system:

$$x_{k+1} = g(x_k) \quad (23)$$

for some given map $g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$. At this point we make no continuity or smoothness assumptions for g . Our main assumption regarding system (23) will be the following:

M: g is non-increasing, i.e. $x_1 \leq x_2 \Rightarrow g(x_1) \geq g(x_2)$

A *nontrivial 2-periodic point* of system (23) is a number $a \in \mathbb{R}_+$ such that $g(a) = b$ for some $b \in \mathbb{R}_+$ with $b \neq a$ and such that $g(b) = a$. For every integer $i > 1$ we denote $g \circ g \circ \dots \circ g$ (g appears i times in this composition) as g^i . Although the following facts are known, it is hard to give a reference for their proofs. Therefore we include them in the Appendix.

Lemma 6. 1. Suppose that **M** holds. Then for each $x_0 \in \mathbb{R}_+$, there exist $y^1, y^2 \in \mathbb{R}_+$ such that

$$g^{2n}(x_0) \rightarrow y^1 \quad \text{and} \quad g^{2n+1}(x_0) \rightarrow y^2$$

as $n \rightarrow +\infty$, and both convergences are monotonic.

2. Suppose that g is continuous and **M** holds. Then $g(y^1) = y^2$ and $g(y^2) = y^1$, so both y^1 and y^2 are fixed points of g^2 . If g^2 has a unique fixed point y , then y is a globally asymptotically stable fixed point for system (23).

3. Suppose that g is continuous and **M** holds. Then system (23) possesses a unique fixed point $\bar{x} \in \mathbb{R}_+$. Moreover, $\bar{x}$ is globally asymptotically stable if and only if the map g does not possess nontrivial 2-periodic points.

Proof. See Appendix. □

The next result provides a sufficient condition for the unique fixed point to be globally asymptotically stable.

Lemma 7. Suppose that **M** holds and let $\bar{x}$ be the unique fixed point of system (23) in $\mathbb{R}_+$. If the following sector condition holds:

$$\sup_{x \neq \bar{x}} \left| \frac{g(x) - \bar{x}}{x - \bar{x}} \right| < 1 \quad (24)$$

then $\bar{x}$ is a globally asymptotically stable fixed point for system (23).

Proof. The sector condition (24) illustrated in figure 3. If (24) holds, then one may pick $\alpha \in (0, 1)$ with $\alpha > \sup_{x \neq \bar{x}} |(g(x) - \bar{x})/(x - \bar{x})|$ and define a piecewise affine function $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ as follows:

$$h(x) := \begin{cases} -\alpha(x - \bar{x}) + \bar{x} & \text{for } x \in [0, \frac{\alpha+1}{\alpha}\bar{x}] \\ 0 & \text{for } x > \frac{\alpha+1}{\alpha}\bar{x}. \end{cases} \quad (25)$$

The definition of h , assumption **M** and the fact that $\alpha > \sup_{x \neq \bar{x}} |(g(x) - \bar{x})/(x - \bar{x})|$ imply that the following inequalities are satisfied:

$$g(x) \begin{cases} \leq h(x) & \text{for } x \leq \bar{x} \\ \geq h(x) & \text{for } x \geq \bar{x}. \end{cases} \quad (26)$$

By the third part of lemma 6 it suffices to show that system (23) does not possess nontrivial 2-periodic points. Assume that $a \in \mathbb{R}_+$ is a nontrivial 2-periodic point, so obviously $a \neq \bar{x}$. We distinguish two cases.

Case 1: $a < \bar{x}$.

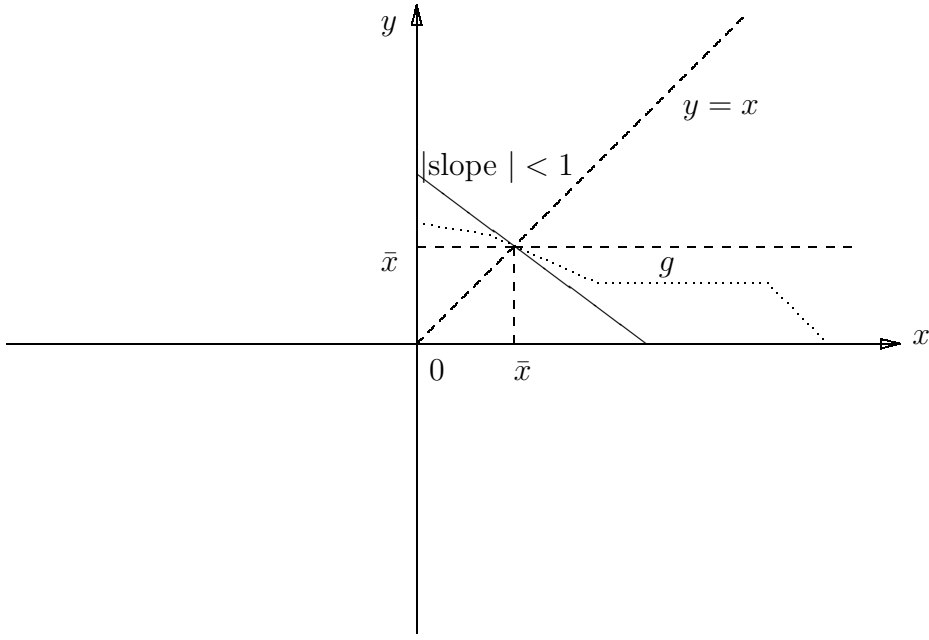


Figure 3: The sector condition (24) for a non-increasing map g .

We will show first that

$$a = (g \circ g)(a) \geq (h \circ h)(a). \quad (27)$$

Since $a < \bar{x}$, it follows from (26) that $g(a) \leq h(a)$ and then **M** implies that

$$a = (g \circ g)(a) \geq (g \circ h)(a). \quad (28)$$

But the definition of h in (25) implies that h is non-increasing also and therefore $a < \bar{x}$ implies that $h(a) \geq h(\bar{x}) = \bar{x}$. Then it follows from (26) that

$$(g \circ h)(a) \geq (h \circ h)(a). \quad (29)$$

From the inequalities (28) and (29) we obtain the desired result (27). Next we will show that

$$(h \circ h)(a) > a. \quad (30)$$

Checking whether (30) holds depends on whether $h(a) \leq$ or $> (\alpha + 1)\bar{x}/\alpha$. If $h(a) \leq (\alpha + 1)\bar{x}/\alpha$ then the definition of h implies $(h \circ h)(a) = (1 - \alpha^2)\bar{x} + \alpha^2 a$ which satisfies (30) since $a < \bar{x}$ and $\alpha \in (0, 1)$. The case $h(a) > (\alpha + 1)\bar{x}/\alpha$ cannot occur because $\alpha \in (0, 1)$ would yield a contradiction to $(\alpha + 1)\bar{x} = h(0) \geq h(a)$ (this last inequality is immediate from the fact that the non-increasing function h must achieve its maximum at $x = 0$). Finally, (27) and (30) yield a contradiction.

Case 2: $a > \bar{x}$.

Define $b = g(a)$ and note that $b < \bar{x}$. Obviously b is also a 2-periodic point of system (23) and the proof reduces to the proof of case 1 if b instead of a is used. $\square$

3 Main results

Recall the system of interest:

$$\begin{pmatrix} \dot{x} \\ \dots \\ \dot{z} \end{pmatrix} = \text{diag}(x, z) \left(\begin{pmatrix} A & \vdots & B \\ \dots & \dots & \dots \\ -C & \vdots & D \end{pmatrix} \begin{pmatrix} x \\ \dots \\ z \end{pmatrix} + \begin{pmatrix} r_1 \\ \dots \\ r_2 \end{pmatrix} \right) \quad (31)$$

where $x \in \mathbb{R}_+^k$ and $z \in \mathbb{R}_+^{(n-k)}$ and for which $\mathbf{H}$ is assumed to hold. Also recall that system (31) can be written as the feedback interconnection of two systems:

$$\dot{z} = \text{diag}(z)(Dz + r_2 + Cu), w = z \quad (32)$$

$$\dot{x} = \text{diag}(x)(Ax + r_1 + Bv), y = x \quad (33)$$

$$v = w \quad (34)$$

$$u = -y \quad (35)$$

where $u \in -\mathbb{R}_+^k$ and $v \in \mathbb{R}_+^{(n-k)}$. Next we summarize some of the properties for this feedback system.

1. Following [1], the I/O system (32) is *monotone* if the state $z \in \mathbb{R}_+^{(n-k)}$, the input $u \in -\mathbb{R}_+^k$ and the output $w \in \mathbb{R}_+^{(n-k)}$ (recall that the orders are the usual partial orders $\preceq$ on the respective spaces); note that the output function is merely the identity on $\mathbb{R}_+^{(n-k)}$. Similarly, the I/O system (33) is monotone if the state $x \in \mathbb{R}_+^k$, the input $v \in \mathbb{R}_+^{(n-k)}$ and the output $y \in \mathbb{R}_+^k$; here the output function is also the identity on $\mathbb{R}_+^k$.
2. By lemma 5, the I/O systems (32) and (33) possess *continuous I/S quasi-characteristics* k_z , respectively k_x (and I/O quasi-characteristics $k_w \equiv k_z$, respectively $k_y \equiv k_x$).
3. By lemma 4 the solutions of system (31) are bounded (in fact, they are uniformly bounded).

Next we state and prove the main result of this paper.

Theorem 3. *If $\mathbf{H}$ holds, then system (31) possesses an almost globally attractive equilibrium point $(\bar{x}, \bar{z}) \in \mathbb{R}_+^n$, provided that the discrete-time system*

$$u_{k+1} = -(k_y \circ k_w)(u_k) \quad (36)$$

which is defined on $-\mathbb{R}_+^k$, possesses a globally attractive fixed point $\bar{u}$. In that case $(\bar{x}, \bar{z}) = ((k_x \circ k_w)(\bar{u}), k_z(\bar{u}))$.

Proof. The theorem is proved if the conditions of theorem 2 are verified. We have already shown that the first 3 conditions (monotonicity, existence of continuous I/S and I/O quasi-characteristics, and boundedness of solutions) are satisfied. The last, small-gain condition holds because of the assumption that (36) possesses a globally attractive fixed point $\bar{u} \in -\mathbb{R}_+^k$. $\square$

Remark 5. Although this is not apparent from the above proof, it can be shown that under the conditions of theorem 3 the zero-measure set of non-convergent initial conditions of system (31) is (a subset of) the boundary of $\mathbb{R}_+^n$. This implies in particular that all solutions initiated in the interior of $\mathbb{R}_+^n$ converge to the equilibrium point $(\bar{x}, \bar{z})$. We refer to [2] for more on this.

Remark 6. Note that no (local) stability information is provided by theorem 3, as only a convergence result is given. However, we shall illustrate below that the small gain condition in theorem 3 may imply local stability of the equilibrium point, by simply checking the stability properties of the linearization at the equilibrium point.

Of course it is in general very hard to determine whether the discrete-time system (36) possesses a globally attractive fixed point. Under an extra, fairly natural condition for system (31), this task may be simplified as we will see below. This condition is the following rank condition:

$$\mathbf{R}: \text{Rank}(B) = \text{Rank}(C) = 1.$$

The biological interpretation of condition $\mathbf{R}$ is that there is no prey-selection by predators, nor does it matter to a prey species by which predator its individuals are eaten.

If $\mathbf{H}$ and $\mathbf{R}$ hold then there exist nonzero vectors $b, \gamma \in \mathbb{R}_+^k$ and $c, \beta \in \mathbb{R}_+^{(n-k)}$ such that $B = b\beta^T$ and $C = c\gamma^T$. (Note that these vectors are not unique since scalar multiples can be

found satisfying the same conditions.) It follows that system (31) can be written as the following feedback interconnection:

$$\dot{z} = \text{diag}(z)(Dz + r_2 + cu), \quad w = \beta^T z \quad (37)$$

$$\dot{x} = \text{diag}(x)(Ax + r_1 + bv), \quad y = \gamma^T x \quad (38)$$

$$v = w \quad (39)$$

$$u = -y \quad (40)$$

where $u \in -\mathbb{R}_+$ and $v \in \mathbb{R}_+$. As before the I/O systems (37) and (38) are monotone and possess continuous I/S and I/O quasi-characteristics k_z, k_x and $k_w \equiv \beta^T k_z, k_y \equiv \gamma^T k_x$ and boundedness of solutions is immediate from lemma 4. Then another straightforward application of theorem 2 yields:

Corollary 1. *If **H** and **R** hold, then system (31), or equivalently the feedback system (37) – (40), possesses an almost globally attractive equilibrium point $(\bar{x}, \bar{z}) \in \mathbb{R}_+^n$, provided that the **scalar** discrete-time system*

$$u_{k+1} = -(k_y \circ k_w)(u_k) \quad (41)$$

which is defined on $-\mathbb{R}_+$, possesses a globally attractive fixed point $\bar{u}$. In that case $(\bar{x}, \bar{z}) = ((k_x \circ \beta^T k_z)(\bar{u}), k_z(\bar{u}))$.

Remark 7. Notice that the small-gain condition in corollary 1 is equivalent to the following small-gain condition:

The scalar discrete-time system

$$\tilde{u}_{k+1} = (k_y \circ k_w)(-\tilde{u}_k) := g(\tilde{u}_k) \quad (42)$$

which is defined on $\mathbb{R}_+$, possesses a globally attractive fixed point $\bar{\tilde{u}}$.

This equivalence follows immediately from the coordinate transformation $\tilde{u}_k = -u_k$. Observe that $g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a scalar and continuous map since k_y and k_w are continuous by lemma 5. Also, remark 3 implies that k_w and k_y are monotone and thus that g is non-increasing (or equivalently, satisfies **M**). But then lemma 7 can be used to verify whether the small-gain condition for system (42), or equivalently for system (41), is satisfied. This may lead to the mentioned simplification as we will illustrate below on both examples from the Introduction.

Remark 8. Both remarks following theorem 3 apply to corollary 1 as well.

Example 1 (continued)

Define $b = (1 \ 0)^T$, $\beta = 1$, $c = k$ and $\gamma = (0 \ 1)^T$ and rewrite system (2) from Example 1 in the form (37)-(40). Using the characterization (21) in lemma 3, the I/O quasi-characteristics k_w and k_y are computable, yielding the following explicit form for system (42):

$$\tilde{u}_{k+1} = \begin{cases} (-\frac{k}{3})\tilde{u}_k + (1 + \frac{k}{3}) & \text{for } \tilde{u}_k \in [0, 1 + \frac{3}{2k}] \\ \frac{1}{2} & \text{for } \tilde{u}_k > 1 + \frac{3}{2k} \end{cases} \quad (43)$$

Since $k > 0$, it is easily verified that system (43) possesses a fixed point $\bar{\tilde{u}}$ in the interval $(0, 1 + \frac{3}{2k})$. Choosing $\alpha \geq 0$ as:

$$\alpha = \frac{k}{3} < 1 \quad (44)$$

we see that the conditions of lemma 7 are satisfied. Notice that the condition (44) is very close to a necessary condition for global asymptotic stability of $\bar{\tilde{u}}$. Indeed, if $\frac{k}{3} > 1$ then $\bar{\tilde{u}}$ is (locally) unstable. Corollary 1 implies that system (2) possesses an almost globally attractive equilibrium point at $(1, 1, 1)^T$, if condition (44) holds. Of course, the small-gain condition (44) also yields that the equilibrium point is locally stable by recalling from Example 1 that $(1, 1, 1)^T$ is locally asymptotically stable if $0 < k < k_c = 57$. Remark 5 implies that the domain of attraction of $(1, 1, 1)^T$ contains the interior of $\mathbb{R}_+^3$. In fact, the interior of $\mathbb{R}_+^3$ is the domain of attraction, since it is not difficult to see that the boundary of $\mathbb{R}_+^3$ is an invariant set for system (3). Finally note that the small-gain gain-condition (44) is very strong compared to the local stability condition

$k < 57$. However, as we have shown, it guarantees the much stronger property of *almost global asymptotic stability*.

Example 2 (continued)

Define $b = (1/2 \ 1)^T$, $\beta = 1$, $c = k$ and $\gamma = (1 \ 1/2)^T$ and rewrite system (3) from Example 2 in the form (37)-(40). The I/O quasi-characteristics k_w and k_y are computable using the characterization (21) in lemma 3, and yield the following explicit form for system (42):

$$\tilde{u}_{k+1} = \begin{cases} (-\frac{5}{3}k)\tilde{u}_k + \frac{5k+3}{2} & \text{for } \tilde{u}_k \in I_1 := [0, \frac{3}{2} + \frac{2}{3k}] \\ (-\frac{1}{6}k)\tilde{u}_k + \frac{k+2}{4} & \text{for } \tilde{u}_k \in I_2 := (\frac{3}{2} + \frac{2}{3k}, \frac{3}{2}(1 + \frac{1}{k})] \\ \frac{1}{4} & \text{for } \tilde{u}_k \in I_3 := (\frac{3}{2}(1 + \frac{1}{k}), +\infty) \end{cases} \quad (45)$$

Since $k > 0$, it is easily verified that system (45) possesses a fixed point $\tilde{u}$ in the interval $(0, \frac{3}{2} + \frac{2}{3k})$. Choosing $\alpha \geq 0$ as:

$$\alpha = \frac{5}{3}k < 1 \quad (46)$$

we see that the conditions of lemma 7 are satisfied since the slope of g on the interval I_1 equals $-5k/3$ which is smaller than the slopes of g on the intervals I_2 and I_3 which equal $-k/6$, respectively 0. Notice that the condition (46) is very close to a necessary condition for global asymptotic stability of $\tilde{u}$. Indeed, if $\frac{5}{3}k > 1$ then $\tilde{u}$ is (locally) unstable. Now it follows from corollary 1 that system (3) possesses an almost globally attractive equilibrium point at $(1, 1, 1)^T$, provided condition (46) holds. Obviously, this small-gain condition (46) also yields that this equilibrium point is locally stable. Indeed, recall from Example 2 that an application of the Routh-Hurwitz criterion showed that $(1, 1, 1)^T$ is locally asymptotically stable if $0 < k < k_c = 105/2$. Remark 5 implies that the domain of attraction of $(1, 1, 1)^T$ contains the interior of $\mathbb{R}_+^3$. In fact, the interior of $\mathbb{R}_+^3$ is the domain of attraction, since the boundary of $\mathbb{R}_+^3$ is easily seen to be an invariant set for system (3). Note that the small-gain gain-condition (46) is very strong compared to the local stability condition $k < 105/2$. However, as we have shown, it guarantees the much stronger property of *almost global asymptotic stability*.

Finally we performed a few simulations to see what happens for k -values in the interval $(3/5, 105/2)$. Using Matlab, the time series of the components of the solutions with initial condition $x(0) = (0.1, 0.1, 0.1)^T$ are plotted for two different k -values in figures 4 and 5. It appears that the solutions converge in an oscillatory manner to the equilibrium point $(1, 1, 1)^T$. This might indicate that for intermediate k -values, the equilibrium point $(1, 1, 1)^T$ is also almost globally asymptotically stable.

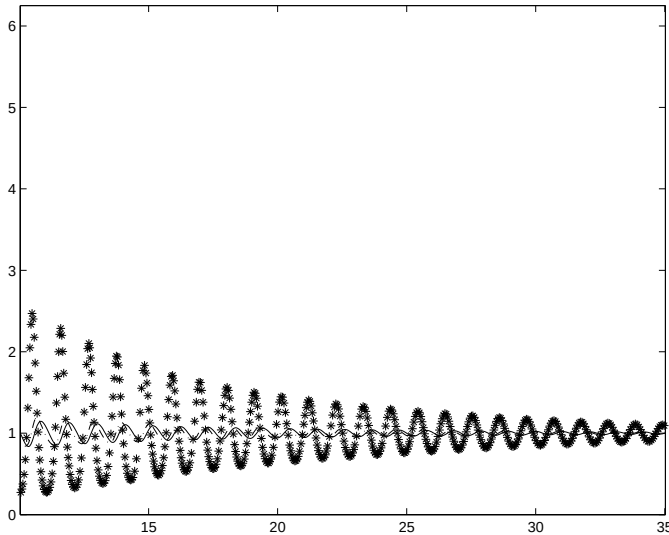


Figure 4: Time series for $k = 30$. $x_1(t)$: '—', $x_2(t)$: '-.-', $z(t)$: '*'.

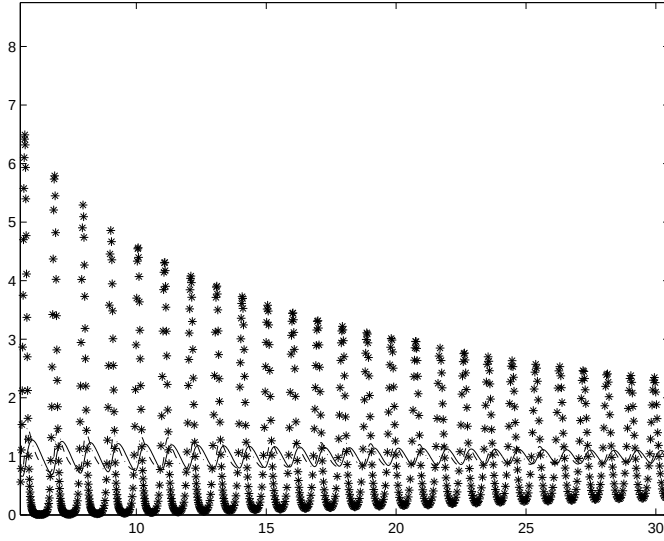


Figure 5: Time series for $k = 46$. $x_1(t)$: '—', $x_2(t)$: '— .', $z(t)$: '*'.

4 Appendix: Proof of Lemma 6

1. Let $p(x) := g^2(x)$. Note that p , and therefore every power p^n too, are nondecreasing. Consider the sequence $x_n := p^n(x_0) = g^{2n}(x_0)$. Since p is bounded (because $p(x) = g(g(x)) \leq g(0)$), the sequence $\{x_n\}$ is bounded. If $x_0 \leq x_1$, then $x_n = p^n(x_0) \leq p^n(x_1) = x_{n+1}$. If $x_0 \geq x_1$, then $x_n = p^n(x_0) \geq p^n(x_1) = x_{n+1}$. Therefore, the sequence $\{x_n\}$ is monotonic. Thus $x_n \rightarrow y^1$ for some y^1 .

The same argument applies to the sequence $z_n := p^n(z_0) = g^{2n+1}(x_0)$, where we defined $z_0 := g(x_0)$, resulting in $z_n \rightarrow y^2$.

2. If g is continuous, then $g^{2n}(x_0) \rightarrow y^1$ implies $g^{2n+1}(x_0) = g(g^{2n}(x_0)) \rightarrow g(y^1)$, so $y^2 = g(y^1)$, and a similar argument shows that $g(y^2) = y^1$. This implies that $g^2(y^i) = y^i$ for $i = 1, 2$.

Thus, if y is the unique fixed point of g^2 , necessarily $y^1(x_0) = y^2(x_0) = y$ for all x_0 , which means that $q^n(x_0) \rightarrow y$, for all x_0 , implying that y is a globally attractive fixed point of system (23). This means that also $p^n(x_0) \rightarrow y$, for all x_0 .

To prove stability of y , consider any interval of the form $[a, b]$ with $a \leq y \leq b$. If $p(a) < a$ then monotonicity of $\{p^n(a)\}$ would imply that $p^n(a)$ converges to a limit l satisfying $l < a$, contradicting $a \leq y$; so $p(a) \geq a$. Similarly, $p(b) \leq b$, implying that the interval $[a, b]$ is invariant under p since p is non-increasing. Next consider the interval $[g(b), g(a)]$, which contains $g(y) = y$; for the same reasons, this interval is invariant under p . Therefore, $[A, B]$ is invariant under g , where $A = \min\{a, g(b)\}$ and $B = \max\{b, g(a)\}$. This proves stability of the fixed point y of system (23).

3. *Existence and uniqueness of a fixed point*

Existence of a fixed point follows from an application of the intermediate value theorem to the (continuous) function $g(x) - x$ restricted to the closed interval $[0, g(0)]$ and using that g is non-increasing. Uniqueness of the fixed point follows from the fact that g is non-increasing. Denote the unique fixed point as $\bar{x}$.

Global asymptotic stability of $\bar{x}$

Necessity of the non-existence of nontrivial 2-periodic points is obvious. To prove sufficiency, note that if there are no nontrivial 2-periodic points for g , then the map g^2 possesses only one fixed point $\bar{x}$. The result now follows from the previous item.

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